Lead Compensator Theory

3.1 What a Lead Compensator Does

The lead compensator has the form:

$$C(s) = K\frac{s+z}{s+p}, \quad \text{where } |p| > |z|$$

The pole is farther from the origin than the zero. This structure adds positive phase (phase lead) to the loop transfer function at frequencies between the zero and the pole. The maximum **phase lead** a single stage can provide is:

$$\phi_{\max} = \arcsin\left(\frac{p-z}{p+z}\right) \times \frac{180^\circ}{\pi}$$

For practical designs, the pole-zero ratio p/z is typically limited to about 10 to 25 (to avoid noise amplification from the high-frequency pole). This constrains the maximum achievable phase lead from a single stage to approximately 55 to 65 degrees.

The Angle Condition

A point s_d lies on the root locus if and only if the angle condition is satisfied:

$$\angle[K \cdot C(s_d) \cdot G(s_d)] = (2k+1) \times 180^\circ, \quad k = 0, \pm 1, \pm 2, \ldots$$

The angle deficiency is the additional phase that the compensator must provide to place s_d on the root locus:

$$\text{Angle Deficiency} = 180^\circ - |\angle G(s_d)|$$

If the angle deficiency exceeds what a single lead stage can provide (approximately 60 to 70 degrees), a single lead compensator cannot place the desired point on the root locus — regardless of how the zero and pole are chosen.

Aircraft Pitch Plant

The same transfer function from Projects 04 and 05 is used throughout this investigation:

$$\boldsymbol{G(s) = \frac{-1.282 + 1.282}{s^3 + 1.935s^2 + 0.987s + 0.179}}$$

Table 1 Aircraft Pitch Plant Parameters

Parameter	Value	Physical Meaning
K (positive convention)	+1.282	Positive pitch for positive elevator
RHP Zero	s = +1	Non-minimum phase — undershoot at all gains
Pole p_1	−1.2679	Short Period mode — fast, heavily damped
Pole $p_{2,3}$	−0.3336 ± 0.1730j	Phugoid mode — slow, lightly damped, dominant
Gain margin	K = 0.46	From Project 05 — tight stability margin

The performance specifications carried forward from Project 05 are:

Table 2 Performance Specifications and Corresponding Pole Requirements

Specification	Requirement	Pole Constraint
Overshoot	< 10%	Damping ratio $\zeta \geq 0.59$
Settling Time	< 10 s	Real part $\sigma \geq 0.4$

The desired dominant pole location selected for compensator design is s_d = −0.5 + 0.6j, which satisfies $\sigma = 0.5 \geq 0.4$ and $\zeta = 0.64 \geq 0.59$.